

Ziyi Sun

Ocean University of China, Qingdao, Shandong, China

Tel: +86 151-6804-9165 | Email: ziyisun85@gmail.com | Profile: <https://github.com/ziyisun85-ops>

Research Interests: Embodied AI, Robot Learning, Reinforcement Learning, Biomimetic Robotics

EDUCATION

College of Engineering, Ocean University of China

*B.Eng. Candidate in Mechanical Design, Manufacturing and Automation
Sep. 2023 to Jun. 2027 Expected*

- **GPA:** 3.7/4.0 • **Expected Graduate Recommendation Ranking:** 1/52 • **Academic Ranking:** 4/56, Top 7%
- **Selected Coursework:** Numerical Methods, Complex Variables and Integral Transforms, Linear Algebra, Probability Theory and Mathematical Statistics, Mechanical Vibration

Research Experience

A Two-Stage Learning Framework for Autonomous Obstacle Avoidance of an Eel-Like Robot(*First Author, Manuscript Under Review*)

- Developed a vision-based robot learning framework combining imitation learning initialization with PPO fine-tuning for autonomous underwater obstacle avoidance.
- Built a MuJoCo simulation environment with simplified underwater dynamics and AHTA-CPG based action abstraction for strongly coupled eel-like locomotion.
- Applied visual domain randomization and validated the policy in 60 real-world water-tank trials, achieving 70% to 90% success rates across different obstacle settings.

• **Keywords:** Robot Learning, Reinforcement Learning, Imitation Learning, MuJoCo, Sim-to-Real, Visual Domain Randomization

Variable-Length Biomimetic Robotic Fish with Multimodal Perception for Deep-Sea Exploration(Project Lead)

- Led the design of a variable-length biomimetic robotic fish for adaptive operation in unstructured underwater environments.
- Designed modular antagonistic continuum units to enable morphology adaptation.
- Developed multimodal data fusion and motion performance evaluation using Kalman filtering and AHP, with responsibilities in decision-system design, mechanical design, and motion simulation.

• **Key Contributions:** Multimodal data fusion, decision-system development, mechanical design, and motion simulation.

Cross-Domain Biomimetic Eel Robot Based on a Wire-Driven Mechanism

Project Lead | Provincial Innovation and Entrepreneurship Training Project

- Designed a cross-domain biomimetic eel robot for damage inspection and early warning of dynamic submarine cables in offshore floating wind power systems.
- Developed a wire-driven biomimetic structure with buoyancy-gravity center regulation for efficient cruising and dual-mode surfacing/diving.
- Integrated visual recognition and Gaussian splatting for underwater cable reconstruction and damage detection; responsible for mechanical design and hydrodynamic analysis.
- **Key Contributions:** Mechanical design and hydrodynamic analysis of the wire-driven biomimetic eel robot.

Selected Engineering Projects

• **Optical Modeling-Based Thickness Estimation of Semiconductor Epitaxial Layers**

Team Leader | CUMCM 2025 National Second Prize

- Built an infrared-interference-based thickness estimation model for SiC/Si epitaxial layers.
- Derived optical path difference and phase relationships under two-beam and multi-beam interference.
- Applied Savitzky-Golay filtering, Fourier analysis, and the transfer matrix method for high-precision thickness inversion and multi-beam interference correction.
- **Key Contributions:** Refraction model derivation, multi-beam interference correction modeling, and paper writing.

High-Stability Duck-Type Wave Energy Converter Based on a Pendulum PTO System

Core Member | National Innovation and Entrepreneurship Training Project

- Optimized a Salter-D0018 duck-type wave energy converter to improve energy absorption efficiency and operational reliability.
- Used AQWA and genetic algorithms to optimize the hull profile and enhance wave energy capture.
- Developed a pendulum-based PTO design and adaptive wave-condition control strategy to improve power generation stability; responsible for control strategy design, mechanical design, and motion simulation.
- **Key Contributions:** Development of adaptive control strategies, mechanical design, and motion simulation.

Modular Adaptive Deep-Sea Resource Collection and Locomotion Platform

Core Member | Faculty-Led Industry Collaboration Project with China Merchants Group

- Designed a modular deep-sea mining vehicle with an active four-track locomotion system for complex seabed terrain.
- Used Fluent to analyze mining-induced plume generation and developed plume suppression structures with a compliant collection mechanism.
- **Key Contributions:** Analysis and design of a compliant collection mechanism

Publications & Patents

- **Paper:** Design of Biomimetic Robotic Eel with Omnidirectional Quadruple-Wire-Driven Body and Controlled Central Pattern Generator, Published in ICCCR, Zhejiang University. **DOI:** [10.1109/ICCCR65461.2025.11072624](https://doi.org/10.1109/ICCCR65461.2025.11072624)
- **Invention Patent:** High-Stability PTO System for Duck-Type Wave Energy Converters, First Student Inventor.
- **Software Copyright:** Underwater Polarized Light Field Prediction Software for Biomimetic Navigation V1.0, China National Copyright Administration, **Registration No.** [2024SR0140235](https://www.cnipa.gov.cn/zwqj/2024SR0140235).

Honors & Awards

- **Shandong Provincial Government Scholarship**
- **First-Class Comprehensive Scholarship**, Ocean University of China, twice
- **Outstanding Student**, Ocean University of China
- **Outstanding Student Cadre**, Ocean University of China
- **National First Prize**, National 3D Digital Innovation Design Competition
- **National Second Prize**, Contemporary Undergraduate Mathematical Contest in Modeling, 2025
- **National Third Prize**, 18th National College Student Energy Conservation and Emission Reduction Social Practice and Science & Technology Competition
- **Honorable Mention**, Mathematical Contest in Modeling (MCM), 2025

Technical Skills

- **Programming:** Python, C
- **Robot Learning:** Reinforcement Learning, Imitation Learning, Sim-to-Real Transfer
- **Simulation & Modeling:** MuJoCo, MATLAB
- **Computer Vision:** Visual Recognition, Gaussian Splatting, Visual Domain Randomization
- **Robotics & Control:** Motion Simulation, Embedded Control, Mechatronic Debugging
- **Mechanical Design:** SolidWorks, 3D Modeling, Structural Design
- **Languages:** **Mandarin Chinese**, native; **English**, CET-6: 482, CET-4: 591; **German**, elementary coursework; **Italian**, elementary with early exposure

Leadership & Service

- **Class Representative**, Ocean University of China
- **Team Leader**, Engineering College Debate Team
- **Director of Event Organization Department**, Student Association for the Study of Socialism with Chinese Characteristics, Ocean University of China
- **Volunteer**, 2023 Qingdao Marathon, Qingdao, China
- **Member**, 2024 National College Student Volunteer Outreach Team for Promoting the Spirit of the Zunyi Conference
- **Outstanding Individual in Summer Social Practice**, Ocean University of China
- **Active Participant in Youth League Activities**, Ocean University of China
- **Science and Technology Innovation Pioneer Team**, College of Engineering, Ocean University of China